Tesselle

PROJECT · MISSIONLOCK

GOOGLE CLOUD AI BUILDER CUP 2026 · RETAIL & COMMERCE

Imagine it.
Keep what matters.
Get it made.

A personal commerce agent that turns any spark - words, a photo, a video, a glance through smart glasses - into a brief it protects all the way to delivery.

project-tesselle.web.app

Scene generated with Gemini



TEAM TESSELLE · ASHFAQUE RIFAYE

01 · THE PROBLEM

Inspiration is easy. Getting the right thing is not.

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Similar isn't suitable

A product that looks right can be 4 cm too deep, need drilling, or miss the one compartment that mattered.

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Search misses aren't answers

When stock doesn't fit, shoppers give up or quietly buy a compromise - and nobody can make the thing they actually wanted.

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Intent gets lost in the middle

Counteroffers, substitutions, quote changes and device switches rewrite requirements without anyone noticing.

The missing piece isn't more products. It's an agent that **preserves what the customer needs** through the messy commercial middle.

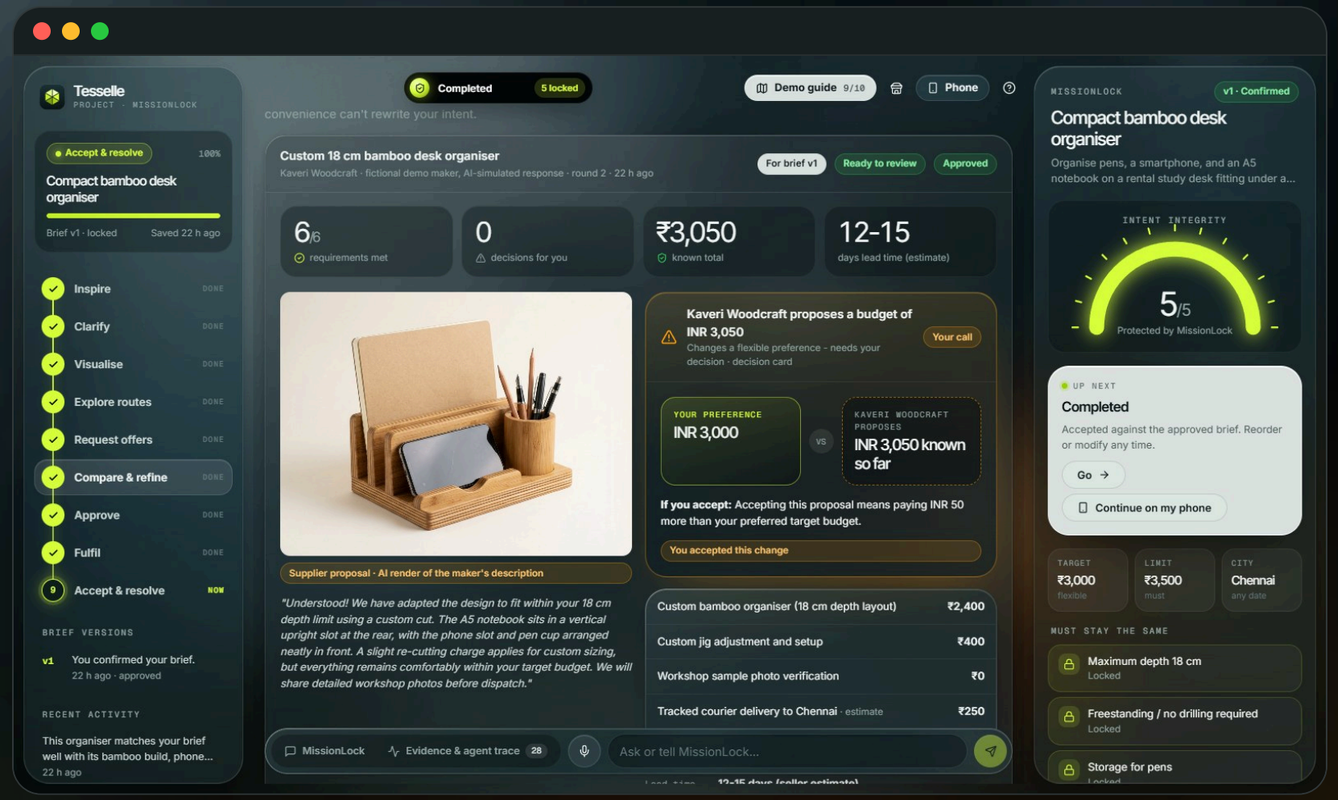
One agent. One brief. Four credible routes.

MissionLock - the intent safeguard

A living, versioned brief: must-haves, flexible preferences, budget, unknowns and acceptance checks. Nothing material changes without the customer's explicit decision.

Fulfilment - buy, source, customise or make

Every route is checked against the same brief, requirement by requirement. When nothing fits, a maker gets a clean brief - and their counteroffer becomes the customer's decision.



Language models propose. Deterministic code decides.

- **Fit verdicts** - meets / possible with a change / not suitable, computed in code from structured requirements.
- **Quote deviations** - every offered spec is diffed against the brief version it answers.
- **Approval gates** - no stale quote, no changed brief, no unmet must-have, no duplicate order.
- **Honest uncertainty** - photo sizes stay "Needs checking"; unknown charges never join a total.

Gemini agents only *explain* verdicts - a cheaper or prettier item that breaks a must-have is never promoted.

Maker proposes a 20 cm jig

Blocked

MUST STAY THE SAME
18 cm max

VS

PROPOSED
~~20 cm~~

Customer keeps 18 cm → the maker revises the layout at **16 cm**.
A small price rise becomes a **flexible decision**

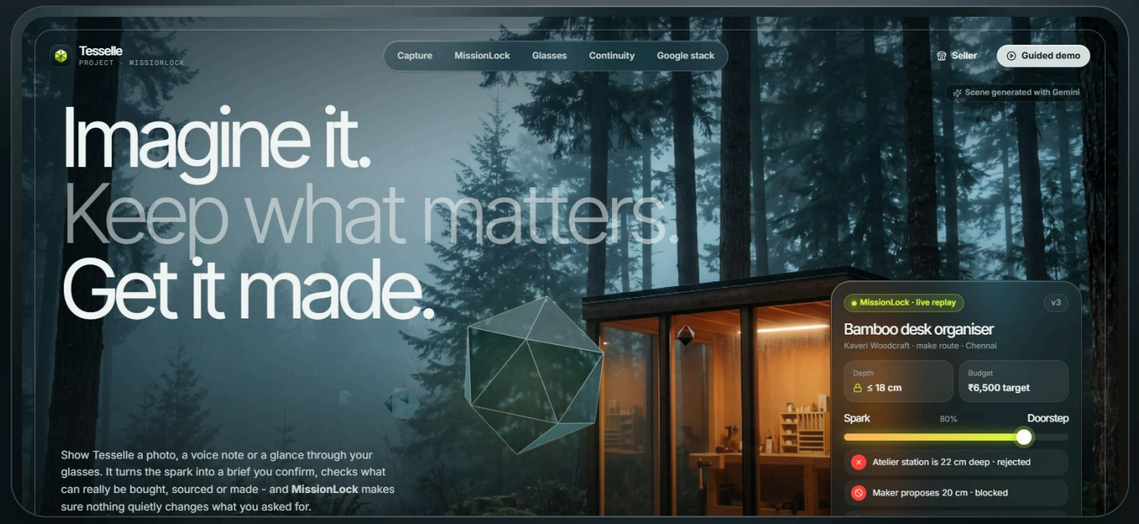
Before that: a gorgeous bamboo station was rejected - **22 cm deep**.

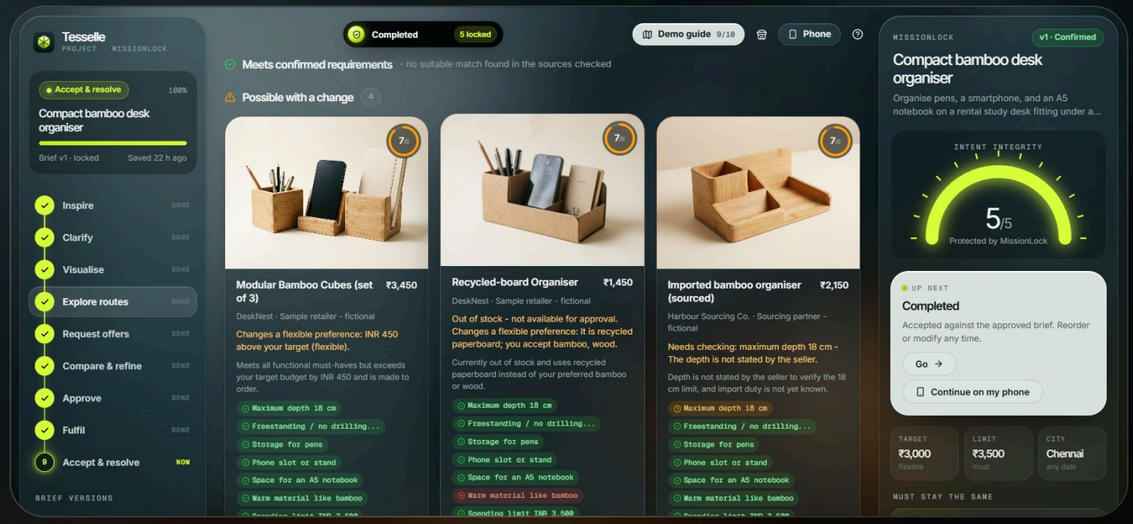
The full journey, tracked live on Google Cloud.

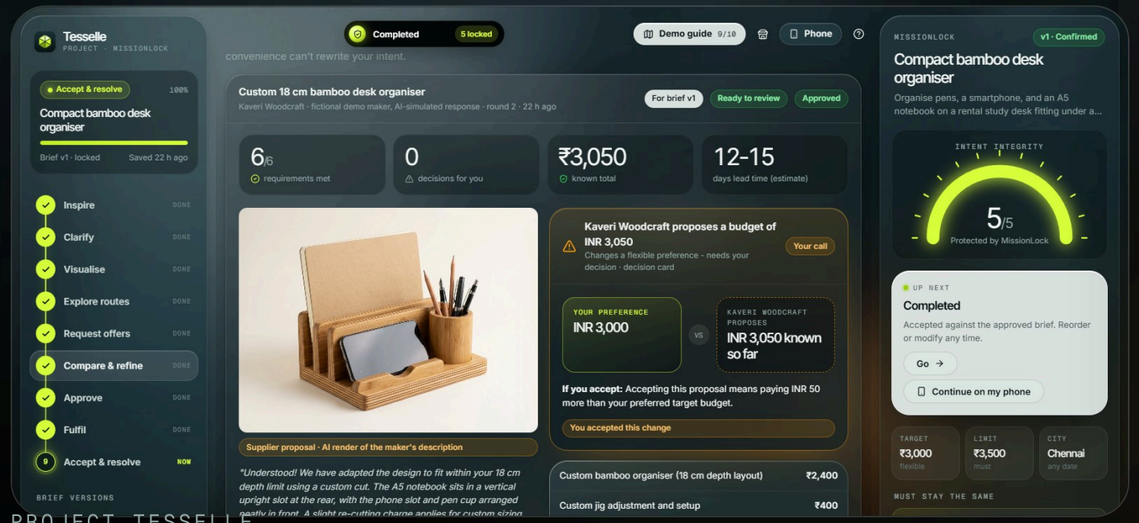
1. **Inspiration**
   Desk photo + "18 cm shelf, I rent, no drilling, about ₹3,000"
2. **Intent**
   Gemini drafts; the customer confirms must-haves vs flexible
3. **Concept**
   Labelled render; pin an area to refine
4. **Existing stock**
   22 cm beauty rejected - honest "no suitable match"
5. **Maker request**
   Privacy-filtered brief, reviewed before sending
6. **Counteroffer**
   20 cm proposal blocked by MissionLock
7. **Resolution**
   Revised 16 cm layout keeps the must-have
8. **Device handoff**
   QR → same pending decision on the phone, spoken recap
9. **Approval**
   Exact spec + quote + brief version; sandbox order
10. **Aftercare**
    Gemini vision checks delivery against approved checks

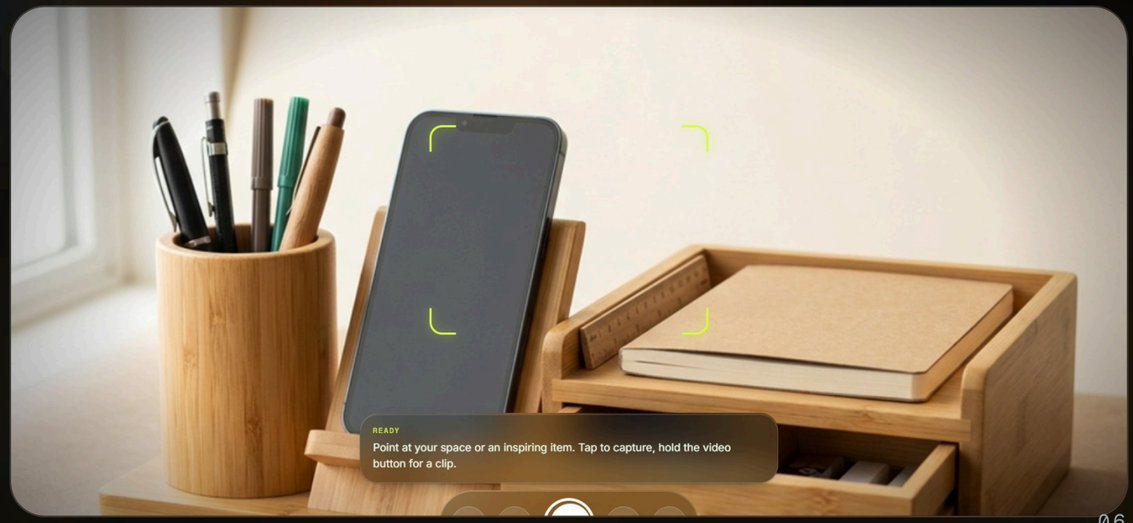
All ten beats are asserted end-to-end against the live deployment by tools/e2e_storyboard.py.

Liquid glass, live tracking - every decision explained.









Nine Gemini agents, one graph workflow, real memory.

INTAKE

Multimodal intake

ADK LlmAgent on Gemini 3.8 Flash with typed output: text, voice, photos, **video** and generated images.

ROUTES

Route explorer

ADK 2 graph Workflow: three scouts in parallel → JoinNode → analyst. Firestore vector search over Gemini embeddings.

STUDIO

Creative studio

Art-director agent + Gemini image generation; click-to-pin visual prompting; flags edits that would touch the brief.

NEGOTIATE

Negotiation

AI-simulated maker agent quotes per brief version; deviation-explainer agent writes decision cards.

MEMORY

Memory & continuity

Agent Platform Sessions per project and **Memory Bank** for durable preferences ("I rent - no drilling").

VOICE + VISION

Voice & vision

Concierge with function tools, Gemini TTS replies, recap agent, and a delivery inspector that knows what a photo can't prove.

07 · ARCHITECTURE

Built end-to-end on Google Cloud.

SURFACES

React liquid-glass workspace + Three.js hero
Phone surface (QR handoff)
Glasses HUD capture
PWA share target (Ray-Ban Meta)
Seller workspace

Firebase Hosting CDN

CLOUD RUN · FASTAPI + GOOGLE ADK 2

9 LlmAgents with output schemas & tools
Graph Workflow (parallel scouts + join)
MissionLock guard - deterministic
NDJSON progress streams with heartbeats
OpenTelemetry → Cloud Trace
Least-privilege service account · no API keys

VERTEX AI / AGENT PLATFORM

Gemini 3.8 Flash (text, vision, video)
Gemini 3.1 Flash Image
Gemini Embedding 2
Gemini 2.5 Flash TTS
Agent Runtime: Sessions + Memory Bank

FIRESTORE

Projects, immutable brief versions, quotes, orders, event trace + vector index

CLOUD STORAGE

Private media: inspirations, concepts, proposals, evidence

CLOUD BUILD + REGISTRY

Container builds, asia-southeast1

LOGGING · TRACE · LABELS

Every Gemini call labelled app=tesselle; daily cost guard

08 · CAPTURE ANYWHERE

From a sentence to a glance through your glasses.

Speak it

Voice notes understood directly by Gemini.

Capture it

Live camera photos and short clips; video understanding.

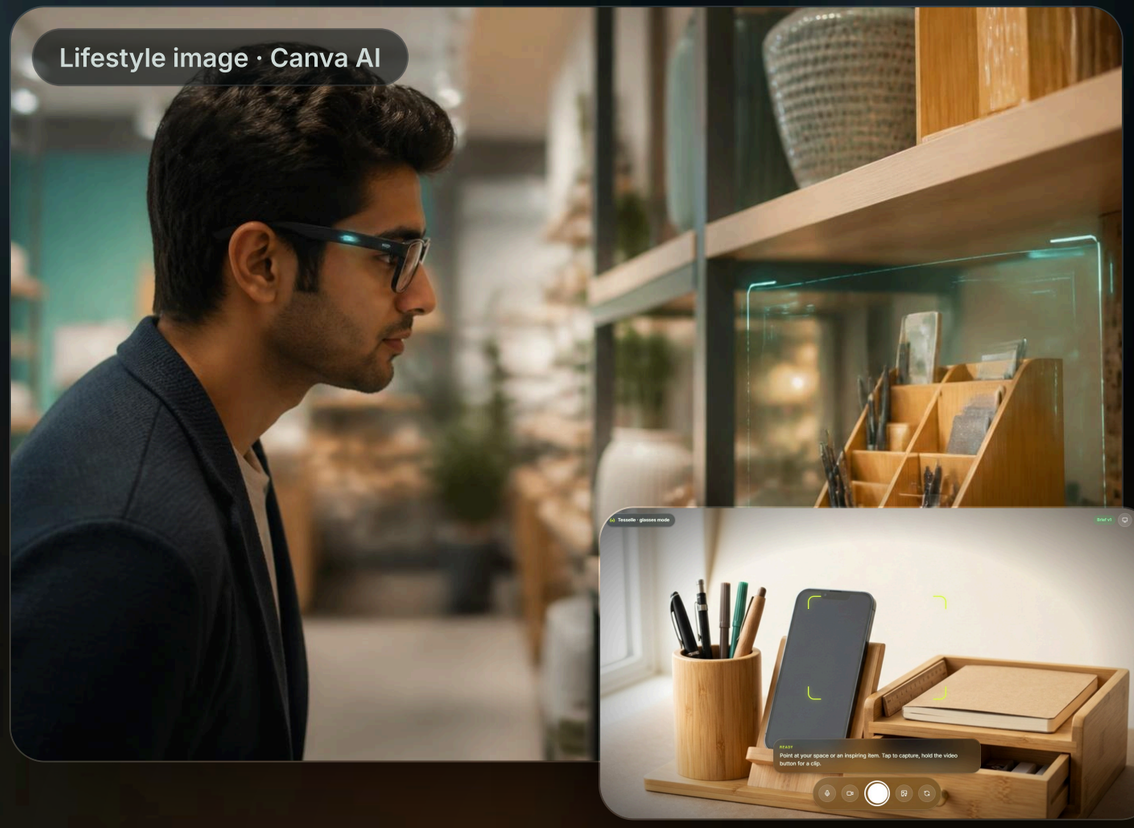
Imagine it

Gemini renders an inspiration, labelled as a concept.

Glance at it

Glasses HUD + PWA share target; replies spoken aloud.

Ray-Ban Meta captures sync to the phone and are shared straight to Tesselle. Direct camera streaming via Meta's native SDK is on the roadmap.



09 · IMPACT & BUSINESS

Unmet demand becomes actionable demand.

Customers

Less repeated searching and explaining, visible trade-offs, access to custom making - and confidence that what arrives is what they approved.

Makers & retailers

Qualified, structured briefs instead of vague enquiries; a way to serve demand existing stock can't. They can decline or propose - never be misrepresented.

Model

Disclosed fee on completed sourcing or custom orders; seller subscriptions for brief and quote management; optional design and sampling services.

What we'll validate (not claimed as achieved)

Brief completion · request-to-quote conversion · suitable-quote acceptance · requirement changes after approval · delivered-item acceptance - against an ordinary search-and-enquiry journey, with consented user testing.

10 · TRUST BY DESIGN

Honest by default. Safe by construction.

- Four unmistakable visual states: concept, existing product, supplier proposal, approved sample.
- Must-haves change only through a labelled decision that creates a new brief version.
- "No suitable match found in the sources checked" - an honest dead end beats a fabricated success.
- Demo sellers are fictional and AI-simulated; orders are sandbox - and labelled so.

- Sellers receive a privacy-filtered brief - no name, address or home imagery unless the customer opts in.
- The conversational agent cannot send requests, accept seller changes or approve spending.
- Vertex AI only, one service identity, no API keys; billing labels on every Gemini call.
- Public-demo cost guard caps daily model and image calls.

From one desk organiser to a demand engine.

NOW

Live prototype

Full storyboard, multimodal + glasses capture, memory, sandbox orders.

NEXT

Real makers

Onboard participating makers in Tamil Nadu and Karnataka; verified quote workflow.

THEN

Live voice & glasses

Gemini Live conversations; Meta wearables SDK for direct capture.

LATER

Shared demand

Opt-in pooling of compatible briefs when minimum quantities block a maker.

Tesselle

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TRY IT

Your intent, kept.

project-tesselle.web.app

tesselle-448771854429.asia-southeast1.run.app

Launch the guided demo - about five minutes, no sign-up. Demo sellers are fictional; orders are sandbox.



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